

A Normal Form Preserving Minimality: An Order-4 Counterexample to the Komori–Kashima Problem

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Abstract

A formula is minimal in a logic L if it belongs to L and no strictly more general formula, with respect to the substitution preorder, belongs to L . Komori and Kashima asked whether every intermediate logic axiomatizable by formulas minimal in classical logic is either intuitionistic or classical logic, and Nakamura and Matsuda gave a counterexample of order 5. We give an explicit axiom of Gödel–Dummett logic of order 4 that is minimal in classical logic, which answers their Open Problem 2; order 4 is the least possible. More generally, over the linear implicative logic **BCI**, a logic is axiomatizable by formulas minimal in classical logic if and only if it is axiomatizable by such formulas of order at most 4. Both results are obtained from a polarized extension-variable translation S that preserves minimality: if α is minimal in a logic L that contains the axiom $a \rightarrow a$ and the formula $S(\alpha)$, then $S(\alpha)$ is also minimal in L . If L contains **BCI**, then $S(\alpha) \in L$ whenever $\alpha \in L$, and $L + \alpha = L + S(\alpha)$. We also show that the results extend to extensions of **FL_e** in the multiplicative language and of **FL_{ew}** in the full language. The main theorem and the soundness of S over **BCI** have been formalized in Lean 4.

Keywords: Intermediate logic, Substructural logic, Minimal formulas, Substitution preorder, Axiomatization, Extension variables.

1 Introduction

We consider implicative formulas, built from variables by $\rightarrow$, which associates to the right. A logic is a set of formulas closed under substitution and modus ponens; for a logic L and a set Γ of formulas, $L + \Gamma$ is the smallest logic containing $L \cup \Gamma$. A formula β is *strictly more general* than a formula α if α is a substitution instance of β but β is not a substitution instance of α . A formula α is *minimal* in a logic L [12] if $\alpha \in L$ and no formula strictly more general than α belongs to L . For example, Peirce’s law $((a \rightarrow b) \rightarrow a) \rightarrow a$ is minimal in classical logic **CL**, whereas $(a \rightarrow b) \rightarrow (a \rightarrow b)$ is not, since it is a substitution instance of the classical tautology $p \rightarrow p$, which is strictly more general. The order of a formula is defined by $\text{Ord}(p) = 1$ for a variable p and $\text{Ord}(\alpha \rightarrow \beta) = \max\{\text{Ord}(\alpha) + 1, \text{Ord}(\beta)\}$.

Komori and Kashima [12, 11] asked which logics are axiomatizable by formulas *minimal* in **CL**, and in particular whether every intermediate logic axiomatizable by such formulas is either intuitionistic logic **IL** or **CL**. Nakamura and Matsuda [13] answered the latter question in the negative: they gave a formula G' of order 5 that is minimal in **CL** and satisfies $\mathbf{IL} + G' = \mathbf{LC}$, where **LC** is Gödel–Dummett logic. They asked whether there exists such a formula of order 4 [13, Open Problem 2].

[Plot: Check the formulation of the problem against [12], [11] and [13, Sect. 1].]

1.1 Main results

We define a map S on formulas. Let α be a formula. Each occurrence of a subformula of α has a polarity, positive or negative: the occurrence of α itself is positive, and in an occurrence of $\beta \rightarrow \gamma$, the occurrence of γ has the same polarity and the occurrence of β the opposite polarity. To each occurrence of a compound subformula of α we assign a new variable, called an *extension variable*, distinct occurrences receiving distinct variables; to each occurrence of a variable p we assign p itself. For an occurrence of a compound subformula $\beta \rightarrow \gamma$ with variable x , where y and z are the variables of the occurrences of β and γ in it, put

$$E = \begin{cases} (y \rightarrow z) \rightarrow x, & \text{if the occurrence is positive,} \\ x \rightarrow (y \rightarrow z), & \text{if the occurrence is negative.} \end{cases}$$

Then $S(\alpha) = E_1 \rightarrow \cdots \rightarrow E_m \rightarrow x_0$, where $E_1, \dots, E_m$ are these formulas for all occurrences of compound subformulas, listed in post-order (for $\beta \rightarrow \gamma$, first those inside β , then those inside γ , then the one for $\beta \rightarrow \gamma$ itself), and x_0 is the variable of α itself; if α is a variable, then $S(\alpha) = \alpha$. If an occurrence of a compound formula γ is the only positive (resp. negative) occurrence of γ in α , we write x_γ^+ (resp. x_γ^-) for its variable; thus $x_0 = x_\alpha^+$. For example, for Peirce's law $\alpha = ((a \rightarrow b) \rightarrow a) \rightarrow a$,

$$S(\alpha) = ((a \rightarrow b) \rightarrow x_{a \rightarrow b}^+) \rightarrow (x_{(a \rightarrow b) \rightarrow a}^- \rightarrow x_{a \rightarrow b}^+ \rightarrow a) \rightarrow ((x_{(a \rightarrow b) \rightarrow a}^- \rightarrow a) \rightarrow x_\alpha^+) \rightarrow x_\alpha^+.$$

Since every E_i has order at most 3, $S(\alpha)$ has order at most 4. (The formal definition is given in Section 3.) Our main theorem is the following.

Theorem 1.1 (Main Theorem). *Let L be a logic containing $I = a \rightarrow a$. If α is minimal in L and $S(\alpha) \in L$, then $S(\alpha)$ is minimal in L .*

Let **BCI** be the linear implicative logic, axiomatized by $B = (b \rightarrow c) \rightarrow (a \rightarrow b) \rightarrow a \rightarrow c$, $C = (a \rightarrow b \rightarrow c) \rightarrow b \rightarrow a \rightarrow c$ and I . If L contains **BCI**, then the hypothesis $S(\alpha) \in L$ follows from $\alpha \in L$, and $L + \alpha = L + S(\alpha)$ for every formula α (Corollary 5.2). We obtain the following two corollaries.

Corollary 1.2. *Let G' be the formula of Nakamura and Matsuda [13, Sect. 4]:*

$$G' = A_1 \rightarrow A_2 \rightarrow \cdots \rightarrow A_7 \rightarrow c,$$

where

$$\begin{aligned} A_1 &= a \rightarrow a', & A_2 &= b' \rightarrow b, & A_3 &= c'' \rightarrow c', & A_4 &= c''' \rightarrow c', \\ A_5 &= (a \rightarrow b) \rightarrow c'', & A_6 &= (b' \rightarrow a') \rightarrow c''', \\ A_7 &= ((a' \rightarrow b') \rightarrow c'') \rightarrow ((b \rightarrow a) \rightarrow c''') \rightarrow c' \rightarrow c, \end{aligned}$$

and put $G'_i = A_i \rightarrow \dots \rightarrow A_7 \rightarrow c$ for $2 \leq i \leq 7$. Then

$$\begin{aligned}
S(G') = & (x_{a \rightarrow a'}^- \rightarrow a \rightarrow a') \rightarrow (x_{b' \rightarrow b}^- \rightarrow b' \rightarrow b) \\
& \rightarrow (x_{c'' \rightarrow c'}^- \rightarrow c'' \rightarrow c') \rightarrow (x_{c''' \rightarrow c'}^- \rightarrow c''' \rightarrow c') \\
& \rightarrow ((a \rightarrow b) \rightarrow x_{a \rightarrow b}^+) \rightarrow (x_{(a \rightarrow b) \rightarrow c''}^- \rightarrow x_{a \rightarrow b}^+ \rightarrow c'') \\
& \rightarrow ((b' \rightarrow a') \rightarrow x_{b' \rightarrow a'}^+) \rightarrow (x_{(b' \rightarrow a') \rightarrow c'''}^- \rightarrow x_{b' \rightarrow a'}^+ \rightarrow c''') \\
& \rightarrow (x_{a' \rightarrow b'}^- \rightarrow a' \rightarrow b') \rightarrow ((x_{a' \rightarrow b'}^- \rightarrow c'') \rightarrow x_{(a' \rightarrow b') \rightarrow c''}^+) \\
& \rightarrow (x_{b \rightarrow a}^- \rightarrow b \rightarrow a) \rightarrow ((x_{b \rightarrow a}^- \rightarrow c''') \rightarrow x_{(b \rightarrow a) \rightarrow c'''}^+) \\
& \rightarrow (x_{c' \rightarrow c}^- \rightarrow c' \rightarrow c) \rightarrow (x_{((b \rightarrow a) \rightarrow c''') \rightarrow c' \rightarrow c}^- \rightarrow x_{(b \rightarrow a) \rightarrow c'''}^+ \rightarrow x_{c' \rightarrow c}^-) \\
& \rightarrow (x_{A_7}^- \rightarrow x_{(a' \rightarrow b') \rightarrow c''}^+ \rightarrow x_{((b \rightarrow a) \rightarrow c''') \rightarrow c' \rightarrow c}^-) \rightarrow ((x_{A_7}^- \rightarrow c) \rightarrow x_{G'_7}^+) \\
& \rightarrow ((x_{(b' \rightarrow a') \rightarrow c'''}^- \rightarrow x_{G'_7}^+) \rightarrow x_{G'_6}^+) \rightarrow ((x_{(a \rightarrow b) \rightarrow c''}^- \rightarrow x_{G'_6}^+) \rightarrow x_{G'_5}^+) \\
& \rightarrow ((x_{c''' \rightarrow c'}^- \rightarrow x_{G'_5}^+) \rightarrow x_{G'_4}^+) \rightarrow ((x_{c'' \rightarrow c'}^- \rightarrow x_{G'_4}^+) \rightarrow x_{G'_3}^+) \\
& \rightarrow ((x_{b' \rightarrow b}^- \rightarrow x_{G'_3}^+) \rightarrow x_{G'_2}^+) \rightarrow ((x_{a \rightarrow a'}^- \rightarrow x_{G'_2}^+) \rightarrow x_{G'}^+) \\
& \rightarrow x_{G'}^+.
\end{aligned} \tag{1}$$

Moreover, $\mathbf{IL} + S(G') = \mathbf{LC}$, the formula $S(G')$ is minimal in $\mathbf{CL}$, and $\text{Ord}(S(G')) = 4$. Hence [13, Open Problem 2] has an affirmative answer.

Every compound subformula of G' occurs exactly once in G' , so the notation x_γ^+ , x_γ^- applies to all variables of $S(G')$ other than those of G' ; the formula (1) contains 30 distinct variables and 67 occurrences of variables. The number 4 in Corollary 1.2 cannot be replaced by 3, since every formula of $\mathbf{CL}$ of order at most 3 belongs to $\mathbf{IL}$ (Proposition 6.1).

Corollary 1.3. *Let L_0 be a logic containing $\mathbf{BCI}$, and let L be a logic. Then $L = L_0 + \Gamma$ for some set Γ of formulas minimal in $\mathbf{CL}$ if and only if $L = L_0 + \Gamma'$ for some set Γ' of formulas of order at most 4 minimal in $\mathbf{CL}$.*

By Proposition 6.1 and Corollary 1.2, the number 4 in Corollary 1.3 cannot be replaced by 3 when $L_0 = \mathbf{IL}$. Corollary 1.3 applies in particular to $L_0 = \mathbf{BCK}$, the base logic of Komori's original formulation of the problem [13, footnote 1].

Theorem 1.1 has been formalized in Lean 4, as have the facts that, for every logic L containing $\mathbf{BCI}$, $S(\alpha) \in L$ for every $\alpha \in L$ and $L + \alpha = L + S(\alpha)$ for every formula α (Appendix B).

1.2 Necessity of the hypotheses

Theorem 1.4 (informal).

- (a) *The hypothesis $I \in L$ cannot be removed from Theorem 1.1: there exist a logic L containing B and C but not I and a formula α minimal in L such that $S(\alpha) \in L$ and $S(\alpha)$ is not minimal in L (Proposition 7.1).*
- (b) *Let $K = a \rightarrow b \rightarrow a$. If L is the logic axiomatized by C and K , or the logic axiomatized by B and K , then K is minimal in L but $S(K) \notin L$; in particular, $L + K \neq L + S(K)$ (Proposition C.1).*

1.3 Substructural logics

Theorem 1.5 (informal).

- (a) *The map S extends to languages containing fusion $\cdot$, $\wedge$, $\vee$ and constants, and Theorem 1.1 holds for this extension (Theorem 8.1).*
- (b) *Let L be a logic containing $\mathbf{FL}_e$ in the language $\{\rightarrow, \cdot, 0, 1\}$, or a logic containing $\mathbf{FL}_{ew}$ in the language $\{\rightarrow, \cdot, \wedge, \vee, 0, 1\}$; the latter includes intermediate logics. Then $L + \alpha = L + S(\alpha)$ for every formula α , and if α is minimal in L , then so is $S(\alpha)$ (Corollary 8.3).*
- (c) *Over $\mathbf{FL}_e$ in the language $\{\rightarrow, \cdot, \wedge, \vee, 0, 1\}$, there is a formula α minimal in $\mathbf{FL}_e$ such that $S(\alpha) \notin \mathbf{FL}_e$, and for every compound formula φ the translation φ^+ of Jeřábek [3, Def. 3.6] is not minimal in any logic containing $\mathbf{FL}_e$ (Remark 8.4).*

1.4 Related work

Extension variables go back to Tseitin [8]. Statman [14] used them to reduce intuitionistic propositional logic to its implicational fragment (see also Švejdar [7]). Plaisted and Greenbaum [5] and Mints [4] use, for each extension variable, only one of the two implications between the variable and the subformula it stands for, chosen according to the polarity of the occurrence. Rybakov [6] uses extension variables in the study of admissible rules.

Jeřábek [3, Def. 3.6] defines a translation $\varphi \mapsto \varphi^+$ of the same kind. It introduces a new variable p_ψ for each occurrence of a subformula ψ of φ that is not a variable, including constants. If $\psi = \psi_0 \circ \psi_1$, then the formula associated with p_ψ is $(p_{\psi_0} \circ p_{\psi_1}) \Rightarrow p_\psi$ or $p_\psi \Rightarrow (p_{\psi_0} \circ p_{\psi_1})$ according to the polarity of the occurrence, where $p_\chi = \chi$ for a variable χ and $\chi \Rightarrow \chi'$ abbreviates $(\chi \rightarrow \chi') \wedge 1$; if ψ is a constant, it is $\psi \Rightarrow p_\psi$ or $p_\psi \Rightarrow \psi$. Finally, φ^+ is the implication from the fusion of these formulas to p_φ . He proves $\mathbf{FL}_e + \varphi = \mathbf{FL}_e + \varphi^+$ for every formula φ [3, Thm. 3.8], and uses this to show that the substructural hierarchy of Ciabattoni, Galatos and Terui [1] collapses to the level $\mathcal{N}_3$ over $\mathbf{FL}_e$. In [3, Example 3.9] he also notes that, for the formula considered there, which contains no lattice connectives, $\rightarrow$ could be used in place of $\Rightarrow$. The map S differs from φ^+ in that it uses $\rightarrow$ in place of $\Rightarrow$, replaces the fusion by an iterated implication, and introduces no variables for constants. Jeřábek does not consider minimality, and φ^+ does not preserve it (Remark 8.4). To our knowledge, Theorem 1.1 has not been observed before.

[Plot: Check the exact forms used in [14], [7], [4].]

1.5 Organization of the paper

Section 2 fixes notation and terminology, Section 3 defines S , and Section 4 proves Theorem 1.1. Section 5 proves Corollary 5.2, and Section 6 proves Corollaries 1.2 and 1.3. Section 7 proves Proposition 7.1, and Section 8 treats substructural logics. Section 9 lists open problems. Appendix A proves Proposition 2.2, Appendix B describes the Lean formalization, and Appendix C proves Proposition C.1.

2 Preliminaries

2.1 Formulas and logics

Except in Section 8, we work in the implicational language: formulas are built from variables by the implication $\rightarrow$. As usual, implication associates to the right: $\alpha \rightarrow \beta \rightarrow \gamma$ stands for $\alpha \rightarrow (\beta \rightarrow \gamma)$. The *order* $\text{Ord}(\alpha)$ of a formula α is defined by $\text{Ord}(p) = 1$ for variables p and $\text{Ord}(\alpha \rightarrow \beta) = \max\{\text{Ord}(\alpha) + 1, \text{Ord}(\beta)\}$, and its *height* by $h(p) = 0$ and $h(\alpha \rightarrow \beta) = 1 + \max\{h(\alpha), h(\beta)\}$.

A *substitution* σ is a map from variables to formulas, and $\sigma\alpha$ denotes the result of applying σ to α . A *logic* is a set of formulas that is closed under modus ponens and substitution. For a set F of formulas, F^* denotes the smallest logic containing F , and for a logic L we write $L + F$ for $(L \cup F)^*$ and $L + \alpha$ for $L + \{\alpha\}$. We shall repeatedly use the following closure properties of a logic L :

- (C1) if $\gamma \in L$, then $\sigma\gamma \in L$ for every substitution σ ;
- (C2) if $\gamma_1 \rightarrow \dots \rightarrow \gamma_k \rightarrow \beta \in L$ and $\gamma_1, \dots, \gamma_k \in L$, then $\beta \in L$;
- (C3) $\gamma \rightarrow \gamma \in L$ for every formula γ .

Every logic satisfies (C1) and (C2), and by (C1) a logic satisfies (C3) if and only if it contains the identity axiom $I = a \rightarrow a$.

2.2 The substitution preorder

We write $\beta \leq_{\text{sub}} \alpha$ if $\alpha = \sigma\beta$ for some substitution σ , and $\beta <_{\text{sub}} \alpha$ if $\beta \leq_{\text{sub}} \alpha$ but not $\alpha \leq_{\text{sub}} \beta$. The relation $\leq_{\text{sub}}$ is the *substitution preorder*. If $\beta <_{\text{sub}} \alpha$, we say that β is *strictly more general* than α .

Definition 2.1. A formula α is *minimal in* a logic L if $\alpha \in L$ and there is no $\beta \in L$ with $\beta <_{\text{sub}} \alpha$.

2.3 Occurrences and a criterion for minimality

We identify an occurrence of a subformula in a formula α with its position in the syntax tree of α , that is, with a finite word over $\{0, 1\}$. The empty word ε denotes the root, and if the subformula at u is $\gamma \rightarrow \delta$, then γ is at $u0$ and δ is at $u1$. We write α_u for the subformula of α at u . An occurrence u is *strictly below* v if v is a proper prefix of u . An occurrence u is a *leaf* if α_u is a variable, and a *non-leaf occurrence* otherwise. Note that two distinct occurrences of the same formula are never strictly below one another.

Let Q be a set of occurrences in α such that no element of Q is strictly below another, and let z be a variable. We write $\alpha[Q:=z]$ for the formula obtained from α by replacing the subformula at each $u \in Q$ by z .

We write $\mathcal{G}(\alpha)$ for the set of formulas $\alpha[Q:=z]$ such that Q is a nonempty set of occurrences of a single formula δ in α , the variable z does not occur in α , and, if δ is a variable, some occurrence of δ does not belong to Q (replacing all occurrences of a variable merely renames it).

Proposition 2.2. *Let L be a logic and let $\alpha \in L$. Then α is minimal in L if and only if $\mathcal{G}(\alpha) \cap L = \emptyset$.*

This is essentially [13, Props. 3.1 and 3.2]; for completeness, we give a proof in Appendix A. [\[Plot: Check whether \[13, Props. 3.1 and 3.2\] cover non-simple formulas.\]](#)

3 The Translation

- ⌈ Fix a formula α . We assign a *polarity*, *positive* or *negative*, to each occurrence in α as follows. The root ε is positive. If $\alpha_u = \alpha_{u0} \rightarrow \alpha_{u1}$, then $u1$ has the same polarity as u and $u0$ the opposite one. For each non-leaf occurrence u we choose a new variable x_u , called an *extension variable*, so that distinct occurrences receive distinct variables and no x_u occurs in α . For every occurrence u let

$$r_u = \begin{cases} \alpha_u, & \text{if } u \text{ is a leaf,} \\ x_u, & \text{if } u \text{ is a non-leaf occurrence.} \end{cases}$$

- ⌈ For every non-leaf occurrence u we put $b_u = r_{u0} \rightarrow r_{u1}$ and

$$E_u = \begin{cases} b_u \rightarrow x_u, & \text{if } u \text{ is positive,} \\ x_u \rightarrow b_u, & \text{if } u \text{ is negative.} \end{cases}$$

- ⌈ **Definition 3.1.** Let $u_1, \dots, u_m$ be the non-leaf occurrences of α , listed in post-order; that is, u_i precedes u_j if and only if u_i is strictly below u_j , or $u_i = w0u'$ and $u_j = w1u''$ for some words w, u', u'' . The *translation* of α is the formula

$$S(\alpha) = E_{u_1} \rightarrow \dots \rightarrow E_{u_m} \rightarrow r_\varepsilon.$$

If α is a variable, then $m = 0$ and $S(\alpha) = \alpha$.

Lemma 3.2. $\text{Ord}(S(\alpha)) \leq 4$ for every formula α .

Proof. Every formula $b_u = r_{u0} \rightarrow r_{u1}$ has order 2, so every formula E_u has order at most 3, and hence $\text{Ord}(S(\alpha)) \leq 4$. $\square$

- ⌈ Let θ_α be the substitution that maps each x_u to α_u and fixes all other variables; then $\theta_\alpha(r_v) = \alpha_v$ for every occurrence v of α .
- ⌈ In examples we use the following notation, similar to that of Jeřábek [3, Def. 3.6]. If u is the only occurrence of a compound formula γ in α that is positive, then we write x_γ^+ for x_u ; if u is the only occurrence of γ in α that is negative, then we write x_γ^- for x_u . In particular, $x_\alpha^+ = x_\varepsilon$.

Example 3.3. For Peirce's law $\alpha = ((a \rightarrow b) \rightarrow a) \rightarrow a$,

$$S(\alpha) = ((a \rightarrow b) \rightarrow x_{a \rightarrow b}^+) \rightarrow (x_{(a \rightarrow b) \rightarrow a}^- \rightarrow x_{a \rightarrow b}^+ \rightarrow a) \rightarrow ((x_{(a \rightarrow b) \rightarrow a}^- \rightarrow a) \rightarrow x_\alpha^+) \rightarrow x_\alpha^+.$$

Example 3.4. For the axiom $\alpha = ((a \rightarrow b) \rightarrow c) \rightarrow ((b \rightarrow a) \rightarrow c) \rightarrow c$ of Gödel–Dummett logic,

$$\begin{aligned} S(\alpha) &= ((a \rightarrow b) \rightarrow x_{a \rightarrow b}^+) \rightarrow (x_{(a \rightarrow b) \rightarrow c}^- \rightarrow x_{a \rightarrow b}^+ \rightarrow c) \\ &\rightarrow ((b \rightarrow a) \rightarrow x_{b \rightarrow a}^+) \rightarrow (x_{(b \rightarrow a) \rightarrow c}^- \rightarrow x_{b \rightarrow a}^+ \rightarrow c) \\ &\rightarrow ((x_{(b \rightarrow a) \rightarrow c}^- \rightarrow c) \rightarrow x_{((b \rightarrow a) \rightarrow c) \rightarrow c}^+) \\ &\rightarrow ((x_{(a \rightarrow b) \rightarrow c}^- \rightarrow x_{((b \rightarrow a) \rightarrow c) \rightarrow c}^+) \rightarrow x_\alpha^+) \rightarrow x_\alpha^+. \end{aligned}$$

4 Proof of the Main Theorem

Throughout this section, α is a formula that is not a variable, $u_1, \dots, u_m$ are its non-leaf occurrences in post-order, and z is a variable not occurring in $S(\alpha)$. By Proposition 2.2, it suffices to show, for a logic L containing I , that if $\mathcal{G}(S(\alpha)) \cap L \neq \emptyset$, then $\mathcal{G}(\alpha) \cap L \neq \emptyset$.

⌈ We first name some occurrences in $S(\alpha)$. For $1 \leq i \leq m+1$, let $T_i = E_{u_i} \rightarrow \dots \rightarrow E_{u_m} \rightarrow x_\varepsilon$; thus $T_1 = S(\alpha)$, $T_{m+1} = x_\varepsilon$, and T_i occurs in $S(\alpha)$ at 1^{i-1} . For $u = u_i$, the formula E_u occurs at $\text{prem}(u) = 1^{i-1}0$, and we let $\text{body}(u)$ and $\text{def}(u)$ be the occurrences of b_u and x_u in E_u ; that is, $\text{body}(u) = \text{prem}(u)0$ and $\text{def}(u) = \text{prem}(u)1$ if u is positive, and $\text{body}(u) = \text{prem}(u)1$ and $\text{def}(u) = \text{prem}(u)0$ if u is negative. Finally, for an occurrence v of α we define the occurrence $\text{rep}(v)$ of r_v in $S(\alpha)$ by $\text{rep}(w0) = \text{body}(w)0$, $\text{rep}(w1) = \text{body}(w)1$ and $\text{rep}(\varepsilon) = 1^m$.

Lemma 4.1.

- (a) Every occurrence in $S(\alpha)$ is exactly one of 1^{i-1} ($1 \leq i \leq m$), $\text{prem}(u)$, $\text{body}(u)$, $\text{def}(u)$ and $\text{rep}(v)$, and the subformula of $S(\alpha)$ there is T_i , E_u , b_u , x_u and r_v , respectively.
- (b) For a non-leaf occurrence u of α , the variable x_u occurs in $S(\alpha)$ exactly at $\text{def}(u)$ and $\text{rep}(u)$. A variable c of α occurs in $S(\alpha)$ exactly at the occurrences $\text{rep}(v)$ with v a leaf of α and $\alpha_v = c$.
- (c) $h(b_u) = 1$, $h(E_u) = 2$ and $h(T_i) \geq 3$ for $i \leq m$. Hence a formula of height 1 occurs in $S(\alpha)$ only at occurrences $\text{body}(u)$, a formula of height 2 only at occurrences $\text{prem}(u)$, and a formula of height at least 3 only at occurrences 1^{i-1} .
- (d) If $b_u = b_v$, then $\alpha_u = \alpha_v$. If $E_u = E_v$, then $u = v$. If $T_i = T_j$, then $i = j$.

Proof. (a) and (b) follow from Definition 3.1, since the extension variables are distinct and do not occur in α .

(c) We have $b_u = r_{u0} \rightarrow r_{u1}$ with r_{u0} and r_{u1} variables, E_u is $b_u \rightarrow x_u$ or $x_u \rightarrow b_u$, and $T_i = E_{u_i} \rightarrow T_{i+1}$. The second claim follows from (a), since the formulas at $\text{def}(u)$ and $\text{rep}(v)$ are variables.

(d) If $b_u = b_v$, then $\alpha_u = \theta_\alpha(b_u) = \theta_\alpha(b_v) = \alpha_v$. Since b_u is not a variable, E_u is of the form $\gamma \rightarrow p$ with p a variable if and only if u is positive. Hence $E_u = E_v$ implies that u and v have the same polarity, so $x_u = x_v$ and $u = v$. Finally, if $i < j$, then T_j is a proper subformula of T_i , so $T_i \neq T_j$. $\square$

⌈ Let $\mathcal{B}(\alpha)$ be the set of formulas $S(\alpha)[P:=z] \in \mathcal{G}(S(\alpha))$ such that every element of P is of the form $\text{body}(u)$ or $\text{rep}(v)$.

Lemma 4.2. Let $\alpha' = S(\alpha)[P:=z] \in \mathcal{B}(\alpha)$, and let

$$Q = \{v \mid \text{rep}(v) \in P\} \cup \{u \mid \text{body}(u) \in P\}.$$

Then $\beta = \alpha[Q:=z] \in \mathcal{G}(\alpha)$, and there are a substitution θ and formulas $\gamma_1, \dots, \gamma_m$ such that

$$\theta\alpha' = (\gamma_1 \rightarrow \gamma_1) \rightarrow \dots \rightarrow (\gamma_m \rightarrow \gamma_m) \rightarrow \beta.$$

Proof. Let δ be the formula such that P is a set of occurrences of δ . We first show that $\beta \in \mathcal{G}(\alpha)$. Clearly $Q \neq \emptyset$, and z does not occur in α , since every variable of α occurs in $S(\alpha)$. If δ is a variable of α , then by Lemma 4.1(b), Q is a set of leaves v with $\alpha_v = \delta$, and some such leaf is not in Q , since P does not contain all occurrences of δ . If $\delta = x_w$, then $P = \{\text{rep}(w)\}$ by Lemma 4.1(b), since $\alpha' \in \mathcal{B}(\alpha)$; hence $Q = \{w\}$, and α_w is not a variable. If δ is not a variable, then Q is the set of u with $\text{body}(u) \in P$, so $b_u = \delta$ and $\alpha_u = \theta_\alpha(\delta)$ for all $u \in Q$.

Since $\alpha' \in \mathcal{B}(\alpha)$, no occurrence 1^{i-1} , $\text{prem}(u)$ or $\text{def}(u)$ belongs to P . In particular, for every non-leaf occurrence v of α ,

$$\alpha'_{\text{body}(v)} = \begin{cases} z, & \text{if } \text{body}(v) \in P, \\ \alpha'_{\text{rep}(v0)} \rightarrow \alpha'_{\text{rep}(v1)}, & \text{otherwise,} \end{cases} \quad (2)$$

where $\alpha'_{\text{rep}(vj)}$ is z or r_{vj} for $j = 0, 1$. Let $\theta(p) = p$ for every variable p that is not an extension variable, and let

$$\theta(x_v) = \theta(\alpha'_{\text{body}(v)})$$

for every non-leaf occurrence v of α . By (2), the only extension variables that may occur in $\alpha'_{\text{body}(v)}$ are x_{v0} and x_{v1} , so this defines θ by induction on $h(\alpha_v)$.

We claim that $\theta(\alpha'_{\text{rep}(v)}) = \beta_v$ for every occurrence v of α that is not strictly below an element of Q . Here $\text{rep}(v)$ is an occurrence in α' , since no proper prefix of $\text{rep}(v)$ belongs to P : the proper prefixes are of the form 1^{i-1} , together with $\text{prem}(w)$ and $\text{body}(w)$ if $v = wj$, and in the latter case $\text{body}(w) \notin P$ because $w \notin Q$. We argue by induction on $h(\alpha_v)$. If $\text{rep}(v) \in P$, then $v \in Q$ and both sides are z . Otherwise $\alpha'_{\text{rep}(v)} = r_v$. If v is a leaf, then $v \notin Q$ and both sides are α_v . If v is a non-leaf occurrence and $\text{body}(v) \in P$, then $v \in Q$ and both sides are z . Otherwise $v \notin Q$, so neither $v0$ nor $v1$ is strictly below an element of Q , and by (2) and the induction hypothesis,

$$\theta(x_v) = \theta(\alpha'_{\text{rep}(v0)}) \rightarrow \theta(\alpha'_{\text{rep}(v1)}) = \beta_{v0} \rightarrow \beta_{v1} = \beta_v.$$

Since no occurrence 1^{i-1} belongs to P , we have $\alpha' = \alpha'_{\text{prem}(u_1)} \rightarrow \dots \rightarrow \alpha'_{\text{prem}(u_m)} \rightarrow \alpha'_{\text{rep}(\varepsilon)}$. Since $\text{def}(u_i) \notin P$, the formula $\alpha'_{\text{prem}(u_i)}$ is $\alpha'_{\text{body}(u_i)} \rightarrow x_{u_i}$ or $x_{u_i} \rightarrow \alpha'_{\text{body}(u_i)}$, and θ maps it to $\gamma_i \rightarrow \gamma_i$ with $\gamma_i = \theta(x_{u_i})$. By the claim for $v = \varepsilon$, θ maps $\alpha'_{\text{rep}(\varepsilon)}$ to β . $\square$

Lemma 4.3. If $\mathcal{G}(S(\alpha)) \cap L \neq \emptyset$ for a logic L , then $\mathcal{B}(\alpha) \cap L \neq \emptyset$.

Proof. Let $\alpha' = S(\alpha)[P:=z] \in L$, where P is a set of occurrences of a formula δ . Since $\sigma\alpha' \in L$ for every substitution σ by (C1), it suffices to find σ such that $\sigma\alpha' \in \mathcal{B}(\alpha)$. We distinguish cases on δ .

If δ is a variable of α or $h(\delta) = 1$, then every element of P is of the form $\text{rep}(v)$ or $\text{body}(u)$, respectively, by Lemma 4.1(b) and (c), so $\alpha' \in \mathcal{B}(\alpha)$.

If $\delta = x_u$, then $P = \{\text{rep}(u)\}$ or $P = \{\text{def}(u)\}$ by Lemma 4.1(b), since P does not contain all occurrences of δ . In the first case $\alpha' \in \mathcal{B}(\alpha)$. In the second case, the substitution that interchanges x_u and z maps α' to $S(\alpha)[\{\text{rep}(u)\}:=z]$.

If $h(\delta) = 2$, then $\delta = E_u$ and $P = \{\text{prem}(u)\}$ for some u by Lemma 4.1(c) and (d). Let $d \in \{0, 1\}$ be such that $\text{def}(u) = \text{prem}(u)d$. Substituting $E_u[\{d\}:=z]$ for z , we obtain $S(\alpha)[\{\text{def}(u)\}:=z]$, which is treated in the previous case.

If $h(\delta) \geq 3$, then $\delta = T_i$ and $P = \{1^{i-1}\}$ for some $i \leq m$ by Lemma 4.1(c) and (d). Substituting $T_i[\{1^{m-i+1}\}:=z]$ for z , we obtain $S(\alpha)[\{\text{rep}(\varepsilon)\}:=z] \in \mathcal{B}(\alpha)$. $\square$

Proof of Theorem 1.1. If α is a variable, then $S(\alpha) = \alpha$ and there is nothing to prove. Otherwise, suppose that $\mathcal{G}(S(\alpha)) \cap L \neq \emptyset$. By Lemma 4.3, there is $\alpha' \in \mathcal{B}(\alpha) \cap L$. Take β, θ and $\gamma_1, \dots, \gamma_m$ as in Lemma 4.2. Then $\theta\alpha' \in L$ by (C1) and $\gamma_i \rightarrow \gamma_i \in L$ by (C3), so $\beta \in L$ by (C2). This contradicts the minimality of α by Proposition 2.2, since $\beta \in \mathcal{G}(\alpha)$. Hence $\mathcal{G}(S(\alpha)) \cap L = \emptyset$, and since $S(\alpha) \in L$, $S(\alpha)$ is minimal in L by Proposition 2.2. $\square$

Example 4.4. Let α be Peirce's law as in Example 3.3, and replace the occurrence of $x_{a \rightarrow b}^+$ in the antecedent $x_{(a \rightarrow b) \rightarrow a}^- \rightarrow x_{a \rightarrow b}^+ \rightarrow a$, that is, at $\text{rep}(00)$, by z :

$$\alpha' = ((a \rightarrow b) \rightarrow x_{a \rightarrow b}^+) \rightarrow (x_{(a \rightarrow b) \rightarrow a}^- \rightarrow z \rightarrow a) \rightarrow ((x_{(a \rightarrow b) \rightarrow a}^- \rightarrow a) \rightarrow x_\alpha^+) \rightarrow x_\alpha^+.$$

Then $Q = \{00\}$ and $\beta = (z \rightarrow a) \rightarrow a$. The substitution θ maps $x_{a \rightarrow b}^+$ to $a \rightarrow b$, $x_{(a \rightarrow b) \rightarrow a}^-$ to $z \rightarrow a$ and x_α^+ to $(z \rightarrow a) \rightarrow a = \beta$, so that

$$\theta\alpha' = ((a \rightarrow b) \rightarrow (a \rightarrow b)) \rightarrow ((z \rightarrow a) \rightarrow (z \rightarrow a)) \rightarrow (\beta \rightarrow \beta) \rightarrow \beta.$$

Since β is not a classical tautology, neither is α' .

5 Soundness over BCI

$\sqsupset$ We now apply Theorem 1.1 to logics containing **BCI**. We use the axiom schemes

$$\begin{aligned} B &= (b \rightarrow c) \rightarrow (a \rightarrow b) \rightarrow a \rightarrow c, & B' &= (a \rightarrow b) \rightarrow (b \rightarrow c) \rightarrow a \rightarrow c, \\ C &= (a \rightarrow b \rightarrow c) \rightarrow b \rightarrow a \rightarrow c, & S &= (a \rightarrow b \rightarrow c) \rightarrow (a \rightarrow b) \rightarrow a \rightarrow c, \\ W &= (a \rightarrow a \rightarrow b) \rightarrow a \rightarrow b, & K &= a \rightarrow b \rightarrow a, \end{aligned}$$

and the logics $\mathbf{BCI} = \{B, C, I\}^*$, $\mathbf{BCK} = \{B, C, K\}^*$, $\mathbf{IL} = \{K, S\}^*$, $\mathbf{CL} = \mathbf{IL} + ((a \rightarrow b) \rightarrow a) \rightarrow a$ and Gödel–Dummett logic [9] $\mathbf{LC} = \mathbf{IL} + ((a \rightarrow b) \rightarrow c) \rightarrow ((b \rightarrow a) \rightarrow c) \rightarrow c$. A formula belongs to $\mathbf{CL}$ if and only if it is true under every Boolean valuation.

[Plot: Check that the implicational axiom of **LC** agrees with [13].]

Lemma 5.1 (Soundness). *If $L \supseteq \mathbf{BCI}$, then $\alpha \rightarrow S(\alpha) \in L$ for every formula α .*

Proof. We argue in the implicational fragment of the sequent calculus for $\mathbf{FL}_e$, which is complete for **BCI** (see [2]). For a non-leaf occurrence u , let Γ_u be the multiset of the formulas E_v such that $v = u$ or v is strictly below u ; for a leaf u , let Γ_u be empty. We show, by induction from the leaves upward, that

$$\Gamma_u, \alpha_u \vdash r_u \quad \text{if } u \text{ is positive,} \quad \Gamma_u, r_u \vdash \alpha_u \quad \text{if } u \text{ is negative.}$$

For leaves, these sequents are axioms. Suppose that u is a positive non-leaf occurrence. Then $u0$ is negative and $u1$ is positive, so the induction hypotheses and the rule $(\rightarrow L)$ yield $\Gamma_{u0}, \Gamma_{u1}, \alpha_u, r_{u0} \vdash r_{u1}$. Applying $(\rightarrow R)$ and then $(\rightarrow L)$ with $E_u = (r_{u0} \rightarrow r_{u1}) \rightarrow x_u$, we obtain $\Gamma_u, \alpha_u \vdash x_u$. Negative occurrences are treated symmetrically. Every formula of Γ_u is used exactly once, so no structural rule other than exchange is needed. Finally, taking $u = \varepsilon$ gives $E_{u1}, \dots, E_{u_m}, \alpha \vdash r_\varepsilon$, and hence $\vdash \alpha \rightarrow S(\alpha)$ by exchange and $(\rightarrow R)$. $\square$

Corollary 5.2. *Let L be a logic containing **BCI**, such as **BCI**, **BCK**, **IL**, **LC** or **CL**. Then $L + \alpha = L + S(\alpha)$ for every formula α . If, moreover, α is minimal in L , then so is $S(\alpha)$.*

Proof. By Lemma 5.1, $S(\alpha) \in L + \alpha$. Conversely, the substitution θ_α of Section 3 maps every formula E_u to $\alpha_u \rightarrow \alpha_u$ and r_ε to α , so $\alpha \in L + S(\alpha)$ by (C1)–(C3). If α is minimal in L , then $\alpha \in L$, hence $S(\alpha) \in L$ by Lemma 5.1, and Theorem 1.1 applies. $\square$

6 Application to the Komori–Kashima Problem

In this section we prove Corollaries 1.2 and 1.3. We first observe that order 3 is not enough.

6.1 Formulas of order at most 3

Proposition 6.1. *Every formula of **CL** of order at most 3 belongs to **IL**.*

Proof. A formula of order at most 2 has the form $p_1 \rightarrow \dots \rightarrow p_k \rightarrow q$ with variables $p_1, \dots, p_k, q$, and we read it as the rule “from $p_1, \dots, p_k$ infer q ”. A formula of order at most 3 has the form $F = H_1 \rightarrow \dots \rightarrow H_n \rightarrow r$, where the H_i are of order at most 2 and r is a variable. Let V be the closure of the empty set under the rules $H_1, \dots, H_n$. The valuation that makes exactly the variables in V true satisfies every H_i , so $r \in V$ because $F \in \mathbf{CL}$. Following the construction of V step by step, we can derive r from $H_1, \dots, H_n$ by modus ponens in **IL**. Hence $F \in \mathbf{IL}$. $\square$

Hence a classical tautology of order at most 3 never axiomatizes a proper extension of **IL**.

6.2 An axiom of Gödel–Dummett logic of order 4

Proof of Corollary 1.2. The formula (1) is obtained directly from Definition 3.1. By [13, Lemma 4.2], G' is minimal in **CL** and $\mathbf{IL} + G' = \mathbf{LC}$. By Corollary 5.2 applied to **IL** and to **CL**, we have $\mathbf{IL} + S(G') = \mathbf{IL} + G' = \mathbf{LC}$, and $S(G')$ is minimal in **CL**. We have $\text{Ord}(S(G')) \leq 4$ by Lemma 3.2, and $\text{Ord}(S(G')) \geq 4$ by Proposition 6.1, since $S(G') \in \mathbf{CL} \setminus \mathbf{IL}$. $\square$

[Plot: The Lean file checks that the displayed formula equals `statman Gprime 8` and has order 4. The formula (1) is generated by `paper/gen_SGprime.py` (supplementary material).]

6.3 A minimality-preserving normal form

Corollary 1.3 is the “only if” direction of the following corollary; the “if” direction is trivial.

Corollary 6.2. *Let L_0 be a logic containing **BCI**. If $L = L_0 + \Gamma$, where every $\alpha \in \Gamma$ is minimal in **CL**, then $L = L_0 + \{S(\alpha) \mid \alpha \in \Gamma\}$, and every $S(\alpha)$ is minimal in **CL** and of order at most 4.*

Proof. The first two claims follow from Corollary 5.2 applied to L_0 and to **CL**, and the last from Lemma 3.2. $\square$

By Proposition 6.1 and Corollary 1.2, the bound 4 cannot be improved. Over **IL**, Corollary 6.2 can be sharpened slightly: since the axioms of order at most 3 may be omitted by Proposition 6.1, a logic axiomatizable over **IL** by **CL**-minimal formulas is either **IL** or axiomatizable over **IL** by **CL**-minimal formulas of order exactly 4.

6.4 Antecedents of order 3

The formula $S(G')$ is rather long. We show that a counterexample of order 4 must have at least two antecedents of order 3, which partly explains this. For a formula $F = A_1 \rightarrow \dots \rightarrow A_n \rightarrow r$ with r a variable, we call $A_1, \dots, A_n$ the *antecedents* of F .

Proposition 6.3. *Let $F = A_1 \rightarrow \dots \rightarrow A_n \rightarrow r \in \mathbf{CL}$, where r is a variable and $\text{Ord}(F) \leq 4$. If at most one A_i has order 3, then either $F \in \mathbf{IL}$ or $\mathbf{IL} + F = \mathbf{CL}$.*

Note that minimality of F is not assumed.

Proof. As in the proof of Proposition 6.1, we call a formula $p_1 \rightarrow \dots \rightarrow p_k \rightarrow q$ with variables $p_1, \dots, p_k, q$ ($k \geq 0$) a *Horn clause* and read it as the rule “from $p_1, \dots, p_k$ infer q ”; these are exactly the formulas of order at most 2. Since $\text{Ord}(F) \leq 4$, every A_i has order at most 3. Recall that the antecedents of a formula can be permuted in **IL**.

Let Γ be the set of antecedents of order at most 2. For a set X of variables, let $\text{Cl}(X)$ be the closure of X under the rules in Γ . We identify a set of variables with the valuation that makes exactly those variables true. Then:

(H1) $\text{Cl}(X)$ satisfies Γ ;

(H2) if $q \in \text{Cl}(X)$, then $\Gamma, X \vdash_{\mathbf{IL}} q$, by modus ponens along the construction of $\text{Cl}(X)$. In particular, if $q \in \text{Cl}(\{p_1, \dots, p_k\})$, then $\Gamma \vdash_{\mathbf{IL}} p_1 \rightarrow \dots \rightarrow p_k \rightarrow q$.

Case 1: no antecedent has order 3. Then F is $\Gamma \rightarrow r$. By (H1) and $F \in \mathbf{CL}$, we have $r \in \text{Cl}(\emptyset)$, and hence $F \in \mathbf{IL}$ by (H2).

Case 2: exactly one antecedent, say $A = B_1 \rightarrow \dots \rightarrow B_m \rightarrow s$, has order 3. Then each B_j has order at most 2. Assume that $F \notin \mathbf{IL}$, and put $V_0 = \text{Cl}(\emptyset)$. We first note two facts.

- $r \notin V_0$, since otherwise $\Gamma \vdash_{\mathbf{IL}} r$ by (H2).
- Some B_j is not derivable from Γ in **IL**. Otherwise $\Gamma, A \vdash_{\mathbf{IL}} s$. Moreover, the valuation $\text{Cl}(\{s\})$ satisfies Γ and A , so $r \in \text{Cl}(\{s\})$ because $F \in \mathbf{CL}$. Then $\Gamma, s \vdash_{\mathbf{IL}} r$ by (H2), and therefore $F \in \mathbf{IL}$, a contradiction.

Fix such a $B_j = p_1 \rightarrow \dots \rightarrow p_k \rightarrow q$, and put $V_1 = \text{Cl}(\{p_1, \dots, p_k\})$. Then $V_0 \subseteq V_1$, and $q \notin V_1$ by (H2). The valuation V_1 falsifies B_j and hence satisfies A ; it also satisfies Γ by (H1). Since $F \in \mathbf{CL}$, we obtain $r \in V_1$.

Let $H_3 = \{0, \frac{1}{2}, 1\}$ be the three-element Gödel algebra, in which $x \Rightarrow y = 1$ if $x \leq y$ and $x \Rightarrow y = y$ otherwise. Define a valuation v by $v(t) = 1$ if $t \in V_0$, $v(t) = \frac{1}{2}$ if $t \in V_1 \setminus V_0$, and $v(t) = 0$ otherwise. A Horn clause $p_1 \rightarrow \dots \rightarrow p_k \rightarrow q$ takes the value 1 under v if and only if $\min_i v(p_i) \leq v(q)$, where the minimum of the empty set is 1. Now:

- every clause in Γ takes the value 1, since V_0 and V_1 are closed under the rules in Γ ;
- B_j takes the value 0, since $v(p_i) \geq \frac{1}{2}$ for all i and $v(q) = 0$; hence A takes the value 1;
- $v(r) = \frac{1}{2}$, since $r \in V_1 \setminus V_0$.

Therefore F takes the value $\frac{1}{2}$, and $H_3 \not\models F$. By Jankov's criterion [10, Theorem 1] (see also [13, Sect. 3.2]), every implicational formula $F \in \mathbf{CL} \setminus \mathbf{IL}$ with $H_3 \not\models F$ satisfies $\mathbf{IL} + F = \mathbf{CL}$. $\square$

[Plot: Check that [10] or [13, Sect. 3.2] covers the implicational fragment.]

Remark 6.4. In $S(\alpha)$, the formula E_u has order 3 if u is positive, since $E_u = b_u \rightarrow x_u$, and order 2 if u is negative, since $E_u = x_u \rightarrow b_u$. Hence the antecedents of order 3 in $S(\alpha)$ are exactly the formulas E_u with u a positive non-leaf occurrence of α . By Proposition 6.3, if $\alpha \in \mathbf{CL}$ has at most one positive implication occurrence, then $\mathbf{IL} + \alpha = \mathbf{IL} + S(\alpha)$ is either $\mathbf{IL}$ or $\mathbf{CL}$. The formula $S(G')$ in (1) has 11 antecedents of order 3, namely the antecedents of the form $(\beta \rightarrow \beta') \rightarrow x_\gamma^+$, where $\gamma \in \{a \rightarrow b, b' \rightarrow a', (a' \rightarrow b') \rightarrow c'', (b \rightarrow a) \rightarrow c''', G'_7, \dots, G'_2, G'\}$, whereas by Proposition 6.3 a $\mathbf{CL}$ -minimal axiom of order 4 of a proper intermediate logic must have at least two antecedents of order 3. We do not know the least possible number.

7 Necessity of the Axiom I

In this section we show that the axiom I cannot be dropped from Theorem 1.1; that B and C cannot be dropped from Corollary 5.2 is shown in Appendix C. The results are not used elsewhere in the paper. We write $\mathbf{BC} = \{B, C\}^*$.

The proof of Theorem 1.1 uses I only to remove the antecedents $\gamma \rightarrow \gamma$ of the formula $\theta\alpha'$ in Lemma 4.2. The next proposition shows that this use is essential, even over $\mathbf{BC}$.

Proposition 7.1. *Let $M = (\{0, 1, 2\}, \rightarrow, \{0\})$ be the matrix with*

$\rightarrow$	0	1	2
0	0	1	1
1	0	0	0
2	0	1	1

and let L_M be the set of formulas that take the value 0 under every valuation in M . Then L_M is a logic containing $\mathbf{BC}$ but not I . Moreover, for $\alpha = (p \rightarrow q) \rightarrow (p \rightarrow q)$ the following hold:

- (a) α is minimal in L_M ;
- (b) $S(\alpha) = (x_{p \rightarrow q}^- \rightarrow (p \rightarrow q)) \rightarrow ((p \rightarrow q) \rightarrow x_{p \rightarrow q}^+) \rightarrow ((x_{p \rightarrow q}^- \rightarrow x_{p \rightarrow q}^+) \rightarrow x_\alpha^+) \rightarrow x_\alpha^+$ belongs to L_M ;
- (c) $S(\alpha)$ is not minimal in L_M , since the formula

$$\alpha' = (x_{p \rightarrow q}^- \rightarrow z) \rightarrow (z \rightarrow x_{p \rightarrow q}^+) \rightarrow ((x_{p \rightarrow q}^- \rightarrow x_{p \rightarrow q}^+) \rightarrow x_\alpha^+) \rightarrow x_\alpha^+,$$

which belongs to $\mathcal{G}(S(\alpha))$ and is obtained by replacing both occurrences of $p \rightarrow q$ by z , belongs to L_M .

Hence Theorem 1.1 fails for logics not containing I , even if $S(\alpha) \in L$.

Proof. The set L_M is closed under substitution, and it is closed under modus ponens because $0 \rightarrow y = 0$ implies $y = 0$. A finite check shows that B and C take only the value 0, whereas $2 \rightarrow 2 = 1$; hence $I \notin L_M$.

(a) We have $\alpha \in L_M$, since the values of implications lie in $\{0, 1\}$ and $0 \rightarrow 0 = 1 \rightarrow 1 = 0$. The elements of $\mathcal{G}(\alpha)$ are

$$(z \rightarrow q) \rightarrow (p \rightarrow q), \quad (p \rightarrow z) \rightarrow (p \rightarrow q), \quad (p \rightarrow q) \rightarrow (z \rightarrow q), \quad (p \rightarrow q) \rightarrow (p \rightarrow z),$$

$$z \rightarrow (p \rightarrow q), \quad (p \rightarrow q) \rightarrow z, \quad z \rightarrow z, \quad z.$$

Each of them takes a value different from 0 under some valuation in M : for $z \rightarrow z$ take $z = 2$, for z take $z = 1$, and for each of the remaining six formulas a refuting valuation is found by a finite check. Hence α is minimal in L_M by Proposition 2.2.

(b) and (c) are verified by finite checks as well. In fact, α' is the type of the linear λ -term $\lambda fgh. h(\lambda y. g(fy))$, and $S(\alpha)$ is its instance under $z \mapsto p \rightarrow q$; both are theorems of **BC**. $\square$

Applying Lemma 4.2 to α' gives $\beta = z \rightarrow z$, and the antecedent of $\theta\alpha'$ that would have to be removed is exactly $z \rightarrow z$, which does not belong to L_M .

[Plot: Checks: [tools/bc_alpha_matrix.py](#) (matrix M , found with z3 and verified by exhaustive evaluation) and [tools/bc_check.py](#) (condensed detachment: $S(\alpha), \alpha' \in \mathbf{BC}$).]

8 Substructural Logics

In this section we extend the translation to richer languages and show that the analogue of Corollary 5.2 holds over **FL_e** in the multiplicative language and over **FL_{ew}** in the full language, but not over **FL_e** with additive connectives. The results of this section are not used elsewhere in the paper.

8.1 The translation in a richer language

Let $\mathcal{L}$ be a language that contains $\rightarrow$ and possibly some of the binary connectives $\cdot$ (fusion), $\wedge$ and $\vee$ and some of the constants $\perp, \top, 0, 1$. An *atom* is a variable or a constant. We consider the full Lambek calculus with exchange **FL_e** and its extension **FL_{ew}** by weakening, in the language $\{\rightarrow, \cdot, \wedge, \vee, 0, 1\}$, possibly expanded by $\perp$ and $\top$ (see [2]). Intuitionistic logic in this language contains **FL_{ew}**. The full Lambek calculus **FL** without exchange has two implications $\backslash$ and $/$ in place of $\rightarrow$. Logics in $\mathcal{L}$, the sets $\mathcal{G}(\alpha)$ and occurrences are defined as in Sections 2 and 4, except that an occurrence u is a leaf if α_u is an atom, and if $\alpha_u = \gamma \circ \delta$ for a binary connective $\circ$, then γ is at $u0$ and δ is at $u1$.

The translation S is defined as in Definition 3.1, with the following changes. If $\alpha_u = \alpha_{u0} \circ \alpha_{u1}$ with $\circ \in \{\cdot, \wedge, \vee\}$, then both $u0$ and $u1$ have the same polarity as u . For a leaf u we put $r_u = \alpha_u$, so that constants are treated like variables, and for a non-leaf occurrence u with $\alpha_u = \alpha_{u0} \circ \alpha_{u1}$ we put $b_u = r_{u0} \circ r_{u1}$.

Theorem 8.1. *Let L be a logic in $\mathcal{L}$ containing I . If α is minimal in L and $S(\alpha) \in L$, then $S(\alpha)$ is minimal in L .*

Proof. The proof of Theorem 1.1 in Section 4 goes through after the following changes:

- variables of α are replaced by **atoms** of α , except in the condition on δ in the definition of $\mathcal{G}(\alpha)$, which still concerns variables only;
- in the “only if” part of the proof of Proposition 2.2 (Appendix A), occurrences of $\rightarrow$ are replaced by occurrences of connectives and constants;
- in Lemmas 4.1(d) and 4.2, the implication in b_u is replaced by the main connective of the corresponding occurrence. $\square$

8.2 Soundness over $\mathbf{FL}_e$ and $\mathbf{FL}_{ew}$

Lemma 8.2. *In each of the following cases, $\alpha \rightarrow S(\alpha) \in L$ for every formula α of $\mathcal{L}$:*

- (i) $\mathcal{L} \subseteq \{\rightarrow, \cdot, 0, 1\}$ and $L \supseteq \mathbf{FL}_e$;
- (ii) $L \supseteq \mathbf{FL}_{ew}$; in particular, $L \supseteq \mathbf{IL}$.

Proof. We argue as in the proof of Lemma 5.1, in the sequent calculus for $\mathbf{FL}_e$, with weakening in case (ii). The case of fusion is similar to that of implication, and again every formula of Γ_u is used exactly once; this proves (i). For $\wedge$ and $\vee$, the rules $(\wedge R)$ and $(\vee L)$ share their context between the two upper sequents, and in each upper sequent we remove the formulas E_v with v in the other subtree by weakening; this proves (ii). $\square$

Corollary 8.3. *In each case of Lemma 8.2, $L + \alpha = L + S(\alpha)$ for every formula α of $\mathcal{L}$. If, moreover, α is minimal in L , then so is $S(\alpha)$.*

Proof. As in the proof of Corollary 5.2, using Lemma 8.2 and Theorem 8.1. $\square$

8.3 Additive connectives over $\mathbf{FL}_e$

The proof of Theorem 8.1 uses only (C1)–(C3), and **polarity** enters only through Lemma 8.2. It is exactly this lemma that fails over $\mathbf{FL}_e$ in the presence of additive connectives. The following two observations, both verified with a cut-free prover for $\mathbf{FL}_e$ (`tools/fl_e_check.py`), show that the difficulty cannot be removed simply by replacing each E_u by $E_u \wedge 1$ as Jeřábek does.

Remark 8.4. Let $\mathcal{L} = \{\rightarrow, \cdot, \wedge, \vee, 0, 1\}$.

- (a) *Without the conjunctions $\wedge 1$, S is not sound over $\mathbf{FL}_e$.* The formula $\alpha = (q \rightarrow q) \wedge (r \rightarrow r)$ is minimal in $\mathbf{FL}_e$, but

$$S(\alpha) = ((q \rightarrow q) \rightarrow x_{q \rightarrow q}^+) \rightarrow ((r \rightarrow r) \rightarrow x_{r \rightarrow r}^+) \rightarrow ((x_{q \rightarrow q}^+ \wedge x_{r \rightarrow r}^+) \rightarrow x_\alpha^+) \rightarrow x_\alpha^+$$

is not provable in $\mathbf{FL}_e$: each upper sequent of the rule $(\wedge R)$ would have to discard one of the antecedents $(q \rightarrow q) \rightarrow x_{q \rightarrow q}^+$ and $(r \rightarrow r) \rightarrow x_{r \rightarrow r}^+$.

- (b) *With the conjunctions $\wedge 1$, minimality is always lost.* Jeřábek’s φ^+ has $E_u \wedge 1$ in place of E_u , which makes it sound over $\mathbf{FL}_e$ [3, Thm. 3.8]. The factor $E_\varepsilon \wedge 1$ is never discarded in the proof, so replacing its conjunct 1 by z gives an element of $\mathcal{G}(\varphi^+)$ that is still provable, by $(\wedge L)$. Hence φ^+ is not **minimal** in any logic $L \supseteq \mathbf{FL}_e$, including $\mathbf{FL}_{ew}$, $\mathbf{IL}$ and $\mathbf{CL}$, whenever φ is compound. For example, for $\varphi = p \rightarrow p$ we have $\varphi^+ = (((p \rightarrow p) \rightarrow x_{p \rightarrow p}^+) \wedge 1) \rightarrow x_{p \rightarrow p}^+$, and $((p \rightarrow p) \rightarrow x_{p \rightarrow p}^+) \wedge z \rightarrow x_{p \rightarrow p}^+$ is provable.

Exhaustive tests over all formulas in $\rightarrow, \wedge, \vee$ with at most four binary connectives and three variables give the following picture. Up to renaming, there are 413 formulas **minimal** in **FL_e**. For 164 of them $S(\alpha)$ is provable, and in all these cases it is **minimal**, as predicted by Theorem 8.1. The formula φ^+ is provable in all 413 cases and **minimal** in none. Replacing E_u by $E_u \wedge 1$ only for occurrences u strictly below an additive occurrence yields a provable formula in all 413 cases, but a **minimal** one in only 255. For $\alpha = p \vee (q \rightarrow q)$, for instance, the conjunction $\wedge 1$ is not needed because the rule $(\vee R)$ selects one disjunct. Thus a minimality-preserving translation over **FL_e**, if it exists, must choose the conjunctions $\wedge 1$ according to the actual proof. We leave open whether such a translation exists, as well as the non-commutative case **FL**.

9 Concluding Remarks

We have shown that a polarized extension-variable translation preserves **minimality** with respect to the **substitution preorder** in every logic that contains the axiom I and the translated formula. As a consequence, axiomatizations of implicational logics over **BCI** by **CL**-minimal formulas can always be taken to consist of formulas of order at most 4, and [13, Open Problem 2] has an affirmative answer. We conclude with some questions that remain open.

1. Which intermediate logics are axiomatizable by **CL**-minimal formulas [13, Open Problem 1]? By Corollary 6.2 and Proposition 6.1, it suffices to consider formulas of order 4.
2. Is there a minimality-preserving translation over **FL_e** with additive connectives, or over the non-commutative calculus **FL** (Section 8.3)?
3. Does the translation, possibly with the formulas E_u listed in a different order, preserve **minimality** over ticket entailment **T_→** (Appendix C)?
4. What is the least number of **antecedents** of order 3 in a **CL**-minimal axiom of order 4 of a proper intermediate logic? By Proposition 6.3 it is at least 2, whereas $S(G')$ has 11 such **antecedents**.

Acknowledgements. [Plot: JSPS KAKENHI etc.]

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[Plot: check]
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A Proof of Proposition 2.2

Proof of Proposition 2.2. “Only if”. Suppose that α is minimal in L . Let $\beta = \alpha[Q:=z] \in \mathcal{G}(\alpha)$; we show that $\beta <_{\text{sub}} \alpha$, so that $\beta \notin L$. Since z does not occur in α , substituting δ for z in β yields α ; hence $\beta \leq_{\text{sub}} \alpha$. Suppose, towards a contradiction, that $\tau\alpha = \beta$ for some substitution τ . Note that a substitution never decreases the number of occurrences of $\rightarrow$.

Case: δ is not a variable. Then δ contains $\rightarrow$, so β has fewer occurrences of $\rightarrow$ than α , which is impossible.

Case: δ is a variable p . Then α and β have the same number of occurrences of $\rightarrow$, so τ maps each variable of α to a variable. Thus $\tau\alpha$ has the same syntax tree as α and carries the variable $\tau(p)$ at every leaf at which α carries p . At these leaves, however, β carries z (at the

elements of Q) as well as p (at an occurrence outside Q). Hence $\tau(p) = z$ and $\tau(p) = p$, a contradiction.

“If”. Suppose that $\mathcal{G}(\alpha) \cap L = \emptyset$. Let $\gamma \in L$ and let σ be a substitution with $\sigma\gamma = \alpha$. We show that $\alpha \leq_{\text{sub}} \gamma$.

Case: σ maps the variables of γ injectively to variables. Then there is a substitution τ with $\tau(\sigma(q)) = q$ for every variable q of γ , and $\tau\alpha = \tau\sigma\gamma = \gamma$.

Case: otherwise. There is a variable q of γ such that either (i) $\sigma(q)$ is not a variable, or (ii) $\sigma(q) = \sigma(q')$ is a variable for some variable $q' \neq q$ of γ . Let z be a variable not occurring in α , and let σ' agree with σ except that $\sigma'(q) = z$. Every occurrence of q in γ gives rise to an occurrence of $\delta := \sigma(q)$ in $\alpha = \sigma\gamma$; let Q be the set of these occurrences. Then $Q \neq \emptyset$ and $\sigma'\gamma = \alpha[Q:=z]$. In case (ii), the occurrences of q' in γ give rise to occurrences of the variable δ that do not belong to Q . Therefore $\sigma'\gamma \in \mathcal{G}(\alpha)$, and $\sigma'\gamma \in L$ by (C1), contrary to the hypothesis. $\square$

B Formalization in Lean

[Plot: Half a page.]

Theorem 1.1, Lemma 5.1, the order bound (Lemma 3.2), Corollary 5.2 and the explicit formula of Corollary 1.2 have been formalized in the Lean 4 proof assistant [15]. The development consists of a single file, `lean/StatmanMinimality.lean`, of about 1300 lines. It uses only the Lean core library, without Mathlib, and contains neither `sorry` nor `native_decide`; the command `#print axioms` reports only the standard axioms `propext`, `Classical.choice` and `Quot.sound`. (The identifiers in the formalization use the name `statman` for the translation S .)

Theorem 1.1 is formalized in exactly the generality stated. A logic is represented by a predicate `Th` on formulas satisfying `PreLogic Th`, which expresses closure under substitution and modus ponens together with all instances $\gamma \rightarrow \gamma$ of I . The main theorem reads

```
statman_minimalIn : PreLogic Th →
  (∀ q, Occ q A → q < N) → MinimalIn Th A →
  Th (statman A N) → MinimalIn Th (statman A N),
```

where the hypothesis on N ensures that the extension variables are fresh. The classical instance `statman_minimal` is derived from it by means of `logic_valid` and `statman_valid`.

Lemma 5.1 and the argument in the proof of Corollary 5.2 are formalized for every logic containing **BCI**, represented by `BCIExt Th` (closure under substitution and modus ponens, and all instances of B , C and I), as

```
BCIExt.statman_imp : Th (A → statman A N),
BCIExt.statman_iff : (∀ q, Occ q A → q < N) →
  (Th A ↔ Th (statman A N)),
```

from which the equation $L + \alpha = L + S(\alpha)$ of Corollary 5.2 follows.

The theorem `BCIExt.statman_minimal` is Corollary 5.2, without further hypotheses. We have also formalized the order bound $\text{Ord}(S(\alpha)) \leq 4$ as `statman_ord`, and the explicit formula (1): `statman_Gprime` states that the formula $S(G')$ in (1), with the variables numbered

as in `paper/gen_SGprime.py`, equals `statman Gprime 8`, and `SGprime_ord` states that it has order 4. The results of Section 8 have not been formalized.

The formal proof follows the paper closely. The lemmas `reduction` and `strict` correspond to Proposition 2.2, and `decode` and `decode_node` correspond to Lemma 4.2; the latter are organized node by node rather than via the reduction to $\mathcal{B}(\alpha)$ in Lemma 4.3.

C Necessity of the Axioms B and C

Theorem 1.1 requires that $S(\alpha) \in L$, which is guaranteed over **BCI** by Lemma 5.1. We show that the axioms B and C cannot be dropped here. We use the logics

$$\mathbf{CK} = \{C, K\}^*, \quad \mathbf{BK} = \{B, K\}^*, \quad \mathbf{BB'KI} = \{B, B', K, I\}^*$$

and ticket entailment $\mathbf{T}_{\rightarrow} = \{B, B', I, W\}^*$.

Proposition C.1. *The following hold.*

(a) (Without B .) The formula K is minimal in **CK**, but

$$S(K) = ((b \rightarrow a) \rightarrow x_{b \rightarrow a}^+) \rightarrow ((a \rightarrow x_{b \rightarrow a}^+) \rightarrow x_K^+) \rightarrow x_K^+$$

does not belong to **CK**.

(b) (Without C .) The formula I is minimal in **BB'KI**, but $S(I) = ((a \rightarrow a) \rightarrow x_I^+) \rightarrow x_I^+$ does not belong to **BB'KI**. Likewise, K is minimal in **BK** and in **BB'KI**, but $S(K) \notin \mathbf{BB'KI} \supseteq \mathbf{BK}$.

Consequently, if L is **CK** or a logic between **BK** and **BB'KI**, then the translation S does not preserve minimality in L , and $L + \alpha \neq L + S(\alpha)$ for $\alpha = K$; for $L = \mathbf{BB'KI}$ the same holds also for $\alpha = I$.

Proof. Minimality. The elements of $\mathcal{G}(K)$ are $z \rightarrow b \rightarrow a$, $a \rightarrow b \rightarrow z$, $a \rightarrow z$ and z , and those of $\mathcal{G}(I)$ are $z \rightarrow a$, $a \rightarrow z$ and z . None of them is classically valid, so none of them belongs to any of the logics under consideration, all of which are contained in **CL**.

(a) Let $M_4 = (\{0, 1, 2, 3\}, \rightarrow, \{0\})$ be the matrix with

$\rightarrow$	0	1	2	3
0	0	1	2	3
1	0	0	1	0
2	0	0	0	3
3	0	1	2	0

A finite check shows that M_4 validates C and K (and hence also I , which is derivable in **CK**) but not B . The set $\{0\}$ is closed under modus ponens, since $0 \rightarrow y = 0$ implies $y = 0$, so every theorem of **CK** takes the value 0. However, under the valuation $a = b = 1$, $x_{b \rightarrow a}^+ = 2$ and $x_K^+ = 3$ we have $b \rightarrow a = 0$, $(b \rightarrow a) \rightarrow x_{b \rightarrow a}^+ = 2$, $a \rightarrow x_{b \rightarrow a}^+ = 1$ and $(a \rightarrow x_{b \rightarrow a}^+) \rightarrow x_K^+ = 0$, so $S(K)$ takes the value $2 \rightarrow (0 \rightarrow 3) = 2 \rightarrow 3 = 3 \neq 0$.

(b) Let $M_3 = (\{0, 1, 2\}, \rightarrow, \{0\})$, where $x \rightarrow y = 0$ if $y \leq x$ and $x \rightarrow y = 1$ otherwise. A finite check shows that M_3 validates B , B' , K and I . As before, $\{0\}$ is closed under modus ponens, so every theorem of **BB'KI** takes the value 0 in M_3 . However, under the valuation $a = 0$ and $x_I^+ = 2$, the formula $S(I)$ takes the value $(0 \rightarrow 2) \rightarrow 2 = 1 \rightarrow 2 = 1$. The formula $S(K)$ is refuted similarly. (The matrix M_3 refutes C and W .) $\square$

Ticket entailment $\mathbf{T}_{\rightarrow}$ contains neither C nor K , and the matrix M_3 does not apply to it, since M_3 refutes W . We do not know whether $S(\alpha) \in \mathbf{T}_{\rightarrow}$ for every $\alpha \in \mathbf{T}_{\rightarrow}$.

[Plot: Checks: `tools/ck_z3.py` (matrix M_4 , found with `z3` and verified by exhaustive evaluation), `tools/bki_matrix.py` (matrix M_3), `tools/ck_cd.py` and `tools/bk_cd.py` (condensed-detachment search, consistent with (a) and (b)). Open: ticket entailment $\mathbf{T}_{\rightarrow}$, where M_3 does not apply because it refutes W ; and whether listing the formulas E_u in a different order helps. For (b), no order helps already for $S(I)$, which has only one such formula.]